BASIC NUMERICAL METHODS (B.COM/ BBA)

Module I- Equations

Equations

An equation is a statement of equality between two expressions. It contains one or more derivatives.

Types of Equations

1. Linear equations
2. Exponential equations
3. Quadratic equations
4. Cubic equations

Properties of Equations

1. Any quantity can be added to both sides.
2. Any quantity can be subtracted from both sides.
3. Any quantity can be multiplied to both sides.
4. Any quantity can be divided to both sides.
5. Any function can be applied to both sides.

Linear equation

It is an equation for a straight line. The power of the unknown in the linear equation will be 1.

Simple linear equation

Simple linear equations are first degree equation in which one variable is unknown.

Simultaneous Equations

Simultaneous equations are a set of equations containing multiple variables.

Quadratic equations

Quadratic equation is a polynomial equation of the second degree. The general form is $ax^2+bx+c=0$.

Factorization

The process of writing an expression as a product of two or more common factor is called method of factorization.

Break Even point

A breakeven point is a point at which there is no profit, there is no loss. Cost are equal to revenue.

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Module II- Matrices

Matrix

Matrix is a rectangular arrangement of numbers or symbols enclosed by a pair of brackets. The term matrix was first introduced in 1850 by J J Sylvester.

Types of Matrices

1. Row matrix

A matrix which has only one row is called a row matrix or row vector.

2. Column matrix

A matrix which has only one column is called a column matrix or column vector.

3. Null matrix

If every element of an m x n matrix is zero, the matrix is called zero matrix or null matrix.

4. Square matrix

If the number of rows and columns in a matrix are same, such matrix is called square matrix or rectangular matrix.

5. Unit matrix

It is a square matrix with 1 on the principal diagonal and zeros elsewhere. It is also called identity matrix.

6. Diagonal matrix

A square matrix in which every non- diagonal elements is zero is called diagonal matrix

7. Scalar matrix

A diagonal matrix whose leading diagonal elements are all equal is called diagonal scalar matrix

8. Upper triangular matrix

A matrix is known as upper triangular matrix if all the elements below the leading diagonal are zero.

9. Lower triangular matrix

A matrix is known as lower triangular matrix if all the elements above the leading diagonal are zero.

10. Sub matrix

The matrix obtained by deleting one or more rows or columns or both of a matrix is called sub matrix.

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11. Equal matrix

Two matrices are called equal matrices if they have the same order or dimension and the corresponding elements are equal.

12. Idempotent matrix

A square matrix is said to be idempotent if $A=A^2$

Singular matrix

A singular matrix is a square one that whose determinant is zero.

Non-singular matrix

A non-singular matrix is a square one whose determinant is not zero.

Transpose of a matrix

The matrix obtained by interchanging rows or columns of a matrix is called transpose.

Adjoint Matrix

Adjoint of a given matrix is the transpose of the matrix formed by co-factors of the elements. It is denoted by Adj A.

Rank of a matrix

The maximum number of its linearly independent columns or rows of a matrix is called rank of a matrix.

Cramer's rule

Cramer's rule is a method for solving linear simultaneous equations. It makes use of determinants and so knowledge of these is necessary before proceeding.

Module III- Sequence, Series and Progression

Sequence

Sequence is an ordered list of an object or events.

Finite sequence

A sequence containing finite number of terms is called finite sequence.

Infinite sequence

A sequence which is not containing finite sequence is called infinite sequence.

Series

The cumulative sum of given sequence of terms is called series.

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Finite series

Finite series means summation of a sequence that has an end.

Infinite series

Infinite series means summation of a sequence that has no end.

Difference between Sequence and Series

Sequence	Series
Set of elements that follows a pattern.	Sum of elements of the sequence.
Order of elements is important.	Order of elements not important
Finite sequence: 1,2,3,4,5	Finite series: 1+2+3+4+5
Infinite sequence: 1, 2, 3, 4.	Infinite series: 1+2+3+4+…….

Progression

A progression is defined as a series of numbers arranged in a predictable manner. It is a type of number set which follows certain definite rules.

Arithmetic progression (AP)

An arithmetic progression is a sequence of numbers such that the difference between the consecutive terms is constant.

Geometric progression (GP)

It is also known as geometric sequence. It is a sequence of numbers where each term after the first is found by multiplying the previous one by a fixed non-zero number called common ratio.

Harmonic progression (HP)

It is defined as a sequence of real numbers which is determined by taking the reciprocals of the arithmetical progression which does not contain zero.

Module IV- Interest and Time value

Simple interest

When the interest is calculated for every period only on the principle then the total interest gained on the all the period is called simple interest.

Compound interest

Compound interest the interest for one period is compounded to the principles to get the principal for the next period.

Annuity

Annuity is a series of equal payments or receipts that occur at spaced intervals.

Present value (PV)

Present value is the current value of future sum of money or stream of cash flows given a specified rate of return.

Effective yield (Effective rate)

It is the actual rate that earn on an investment or pay on a loan after the effects of compounding frequency are considered.

Perpetuity

Perpetuity is an annuity that occurs indefinitely.

Deferred perpetuity

It refers to infinite payment that begin at a later time, because of time value principles of money.

Compounded Annual Growth Rate (CAGR)

CAGR is the rate of return that would be required for an investment to grow from its beginning balance to ending balances.

Equated Monthly Instalments (EMI)

It is a fixed payment amount made by a borrower to a lender at a specified date.

Nominal interest rate

It refers to the interest rate before taking inflation into account. It is the interest rate quoted on bonds and loans.

Real rate of interest

It refers to interest rate after taking inflation into account. This means it adjusts for inflation and gives the real rate of bonds or loan.

Module V- Descriptive Statistics

Desirable qualities of a good average

1. It should be easy to calculate.
2. It should be simple to understand.
3. It should have sample stability.
4. It should be based on all the observations.
5. It should not be affected by average values.

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Measures of central tendency

Average tends to indicate the centre of the distribution is called measures of central tendency.

Types of averages/Measures of central tendency

1. Mean
2. Geometric mean
3. Harmonic mean
4. Median
5. Mode

Arithmetic mean (AM)

The most commonly used measures of central tendency. Mean is the sum of the values divided by the total number of items in the set.

Weighted arithmetic mean

Arithmetic mean computed by considering relative importance of each items is called weighted arithmetic mean.

Merits of AM

- It is easy to calculate.
- It is simple to follow.
- It is finite.
- It is based on the all observations.
- It is rigidly defined.
- It is capable for an algebraic treatment.

Demerits of AM

- It is highly affected by extreme values.
- It cannot be compute accurately.
- It cannot be calculated for qualitative data.
- It is not a suitable average for highly skewed distributions.
- It cannot average the ratios.
- It cannot average the percentage.

Geometric mean (GM)

Geometric mean is the nth positive root of the product of 'n' positive given values.

Merits of GM

- It is rigidly defined.
- It is based on all observations.

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- It is capable for further mathematical treatment.
- It has sampling stability.

Demerits of GM

- It is not simple to understand.
- It is difficult to calculate.
- It cannot be determined graphically.
- It cannot be calculated for open end data.

Harmonic mean (HM)

Harmonic mean is the reciprocal of the arithmetic mean of the reciprocal of the given set of observations.

Merits of HM

- It is rigidly defined.
- It is simple to understand.
- It is easy to calculate.
- It is based on all observations.
- It has sampling stability.
- It is capable for further mathematical treatment.

Demerits of HM

- It is affected by extreme values.
- It cannot be determined graphically.
- It cannot be calculated for open ended data.
- If one of the value is zero, HM cannot be determined.

Median

Median is the middle most observation in a data which is arranged in ascending or descending order.

Merits of median

- It is rigidly defined.
- It is simple to understand.
- It is easy to calculate.
- It can be calculated for open end data.
- It is not much affected by extreme values.
- It can be determined graphically.

Demerits of median

- It is not based on all observation.

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- It is not capable for further mathematical treatment.
- It does not have sampling stability

Mode

Mode is the most frequently occurring value in the data set.

Merits of mode

- It is simple to understand.
- It is easy to calculate.
- It is not much affected by extreme values.
- It can be determined graphically.
- It can be calculated for open end data.
- It has the greatest frequency.

Demerits of mode

- It is not rigidly defined.
- It is not based on all observation.
- It is not capable of further mathematical treatment.
- It does not have sampling stability.

Measures of dispersion

The degree to which numerical data tend to spread about an average value is called dispersion or variation.

Types of measures of dispersion

1. Range
2. Quartile deviation
3. Mean deviation
4. Standard deviation

Range

Range is the difference between highest and lowest values in a given data.

Merits of range

- It is simple to understand.
- It is easy to calculate.
- It is popular measure in weather forecast.
- It is popular measure in the field of medicine.

Demerits of range

- It is not based on all observation.

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- It does not have sampling stability.
- It cannot be calculated for open end data.

Quartile deviation (QD)

Quartiles are three partition values which divides given data into four equal parts. They are denoted Q_1, Q_2, and Q_3.

Merits of QD

- It is rigidly defined.
- It is simple to understand.
- It is easy to calculate.
- It is not unduly affected by extreme values.

Demerits of QD

- It is not based on all observations.
- It does not have sampling ability.
- It is not capable for further mathematical treatment.

Mean deviation (MD)

Mean deviation is defined as the mean of the absolute deviations of observations from some suitable averages like mean median mode.

Merits of MD

It is rigidly defined.
It is based on all observation.
It is less affected by extreme values.

Demerits of MD

- It suffers inaccuracy because sigma of deviation are ignored.
- It cannot be capable for further mathematical treatment.
- It cannot be calculated for open end data.

Standard deviation (SD)

The standard deviation is defined as the positive square root of the mean of the square deviations taken from arithmetic mean of the data.

Merits of SD

- It is rigidly defined.
- It is based on all observation.
- It is capable for further mathematical treatment.
- It has sampling stability.

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Demerits of SD

- It is difficult to understand.
- It is difficult to calculate.
- It cannot be calculated for open end data.

Steps in calculating standard deviation

➢ **Steps for grouped data**

- Take the deviation of the items from assumed mean and denoted it by 'd'
- Multiply the deviations by frequencies.
- Obtain square of the deviation, d^2
- Multiply the squared deviation by frequencies and obtain total
- Substitute the value in the formula

➢ **Step deviation method**

- Find the mid points of various classes
- Take deviations of the mid points from an assumed mean and denote these by 'd'
- Take the common factor and divide the 'd' with it and find 'd'
- Multiply the frequency of the each class.
- Square the deviations and multiply them with the respective frequencies of each class.

Continuous series

Continuous series is a statistical series in which all the class intervals along with their corresponding frequency are listed out in the form of table.

Desecrate series

Desecrate series is a statistical series in which all the observations are listed out along with their corresponding frequency in the form of a table.

Skewness

Lack of symmetry or measure of asymmetry is called skewness. Skewness can be positive or negative.

Positive skewness

If the frequency curve has longer tail to right the distribution is known as positive skewness. Mean> Median> Mode.

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Negative Skewness

If the frequency curve has longer tail to left the distribution is known as negative skewness. Mean< Median< Mode.

Kurtosis

It is statistical measure that calculates the degree of peakedness of a variables probability distribution.

Mesokurtic

If the curve of a frequency distribution is neither too peaked nor too flat then it is called normal curve or mesokurtic.

Platykurtic

If the curve of the frequency distribution is flatter than the mesokurtic curve then it is called platykurtic.

Leptokurtic

If the curve of the frequency distribution is more peaked than the mesokurtic curve then it is called leptokurtic.

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(This is only a short note of the theory part of BASIC NUMERICAL METHODS. For exam preparation, please also refer other available materials including problems according to your syllabus)

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